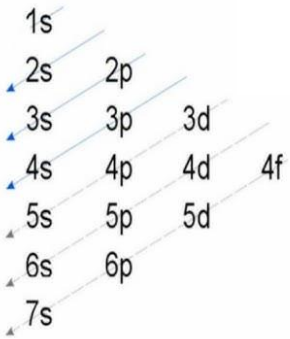
Fe, Fe^{2+}, and Fe^{3+} Electron Configuration

Electronic Configuration

Fe

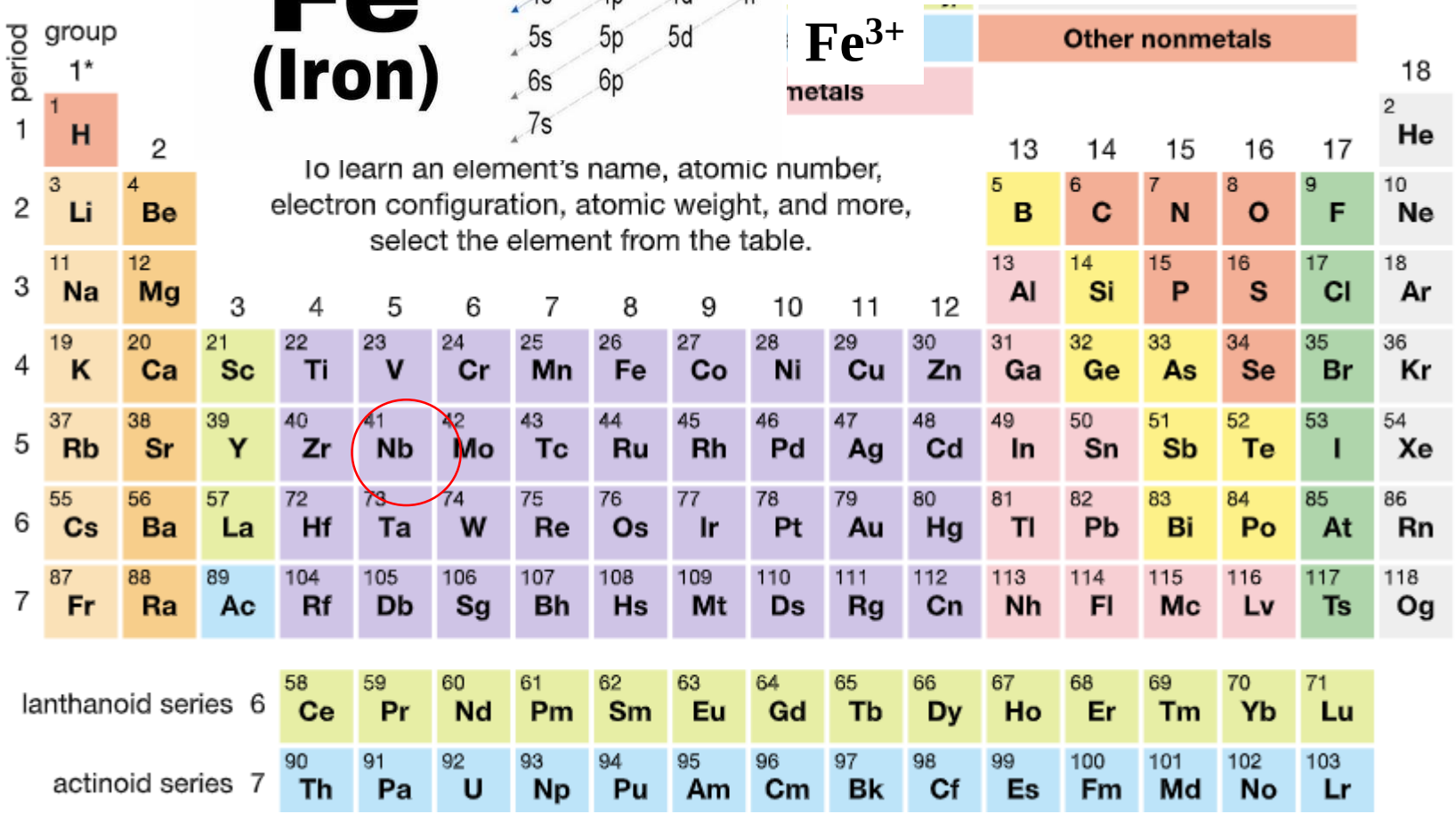
(Iron)



$$Fe = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$$

Fe^{2+}

Fe^{3+}



To learn an element's name, atomic number, electron configuration, atomic weight, and more, select the element from the table.

Home work: Find out the number of d electron in Ni^{2+}; Cr^{3+}; Mn^{2+}; Co^{2+}; Co^{3+}; V^{3+}; Ti^{3+} ions

What is solvation energy?

When a cation or an anion in the gaseous state reacts with a solvent in the liquid state, energy is liberated. This liberated . This liberated energy is termed solvation energy. When the solvent is water, then it is called hydration energy.

Hydration Energies			
Cations		Anions	
Ions	KJ/mol	Ions	KJ/mol
Li^+	-520	F^-	-505
Na^+	-405	Cl^-	-363
K^+	-321	Br^-	-335
Mg^{2+}	-1922	I^-	-298
Ca^{2+}	-1592	NO_3^-	-330
Ba^{2+}	-1305	SO_4^{2-}	-1145
Al^{3+}	-4660	ClO_4^-	-245
Ce^{4+}	-6490	CN^-	-365

1.14 The octet rule

The ground state electronic configurations in Table 1.3 map out a pattern illustrating that filled quantum levels provide 'building blocks' within the electronic configurations of the heavier elements. Worked example 1.6 emphasized that each noble gas is characterized by having a filled quantum level; with the exception of He, this configuration is of the form ns^2np^6. In the early development of bonding models (see *Section 1.11*), the *octet rule* was commonly cited as a means of rationalizing the formation of a particular compound (or ion) which involved an *s*- or *p*-block element. However, the concept of the octet rule is rather limited since it is, strictly, only valid for $n = 2$. Further, many molecules, especially neutral compounds of boron, simply do not contain enough valence electrons for each atom to be associated with eight electrons. Ions (e.g. Mg^{2+} and O^{2-}) often exist only in environments in which electrostatic interaction energies compensate for the energies needed to form the ions from atoms.

An atom obeys the *octet rule* when it gains, loses or shares electrons to give an *outer* shell containing eight electrons with the configuration ns^2np^6.

We have already noted the exception of He, but for $n \geq 3$, there is the possibility of apparently *expanding the octet* (see p. 104). Although the octet rule is still useful at an elementary level, we must bear in mind that it is restricted to a relatively small number of elements. Its extension to the *18-electron rule*, which takes into account the filling of *ns*, *np* and *nd* sub-levels, is discussed in *Sections 20.4* and *23.3*.

Various bonding models

A covalent bond may be defined as a force holding atoms together through sharing of electrons.

The Bohr model of the atom was **proposed by Neil Bohr in 1915**.

Lewis Structures

Gilbert Newton Lewis
1875 – 1946
American Physical Chemist
Discovery: Covalent bonds, concept of electron

- 1916 Proposed the theory of bonding at University of California at Berkeley
- Added information about electrons in the Periodic Table
- Worked on purification of heavy water
- Proposed acid-base theory (Lewis acid and Lewis base concept)
- Worked in the field of photochemistry.
- Nominated 41 times for Nobel prize
- Found dead on March 23 1946 in his laboratory
- At that time he was working with hydrogen cyanide

From NPTL lectures: MS Balakrishna

Lewis structure model

- Determine how the atoms are bonded in a molecule.
- Identify the central atom
- Commonly, H is peripheral, bonded to N, O, S..
- Count the valence electrons of all atoms in the molecule.
- First establish the bonds between the central atom and peripheral atoms with a pair of electrons each.
- Place the remaining electrons around the atoms to complete their octet; begin with the most electronegative atom.
- If needed use double bonds to satisfy the octet.

From NPTL lectures: MS Balakrishna

Failure of the Lewis model

A number of molecules with odd number of electrons exist (no octet) e.g. NO

An atom may not have enough electrons to complete its octet without having formal charges e.g. compounds of group 13 elements with s^2p^1 configuration, BF_3

A central atom having more than 8 electrons

e.g. SF_6

O_2 is paramagnetic!!

Geometry and shapes of the molecules: No information

From NPTL lectures: MS Balakrishna

Lewis model

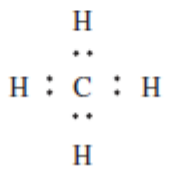
In which of the following covalent compounds is the central atom obeying the octet rule: (a) CH_4; (b) H_2S; (c) ClF_3?

Self-study exercises

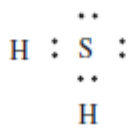
1. Show that N in NF_3 obeys the octet rule.
2. Show that Se in H_2Se obeys the octet rule.
3. In which of the following molecules is the octet rule apparently violated by the central atom: (a) H_2S; (b) HCN; (c) SO_2; (d) CO_2; (e) SO_3? [*Ans.* (c); (e)]
4. Within the series of fluorides IF, IF_3 and IF_5, show that the octet rule appears to be obeyed in only one instance for iodine.

(5) Between $[ClO_2]^-$ and $[ClO_2]$ which is paramagnetic?

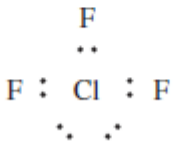
(a) CH_4: A C atom has four valence electrons and forms four covalent bonds by sharing electrons with four H atoms to give an octet. This can be represented by the Lewis structure:



(b) H_2S: An S atom has six valence electrons and forms two covalent bonds by sharing electrons with two H atoms to give an octet. The appropriate Lewis structure which shows that S obeys the octet rule in H_2S is:



(c) ClF_3: A Cl atom has seven valence electrons and should form only one single covalent bond to obtain an octet (e.g. as in Cl_2). In ClF_3, the octet appears to be expanded:



(Valence-shell electron-pair repulsion theory) VSEPR theory: predicting molecular geometries.

farther from the central element A. Therefore the repul...
electron pairs on A decreases

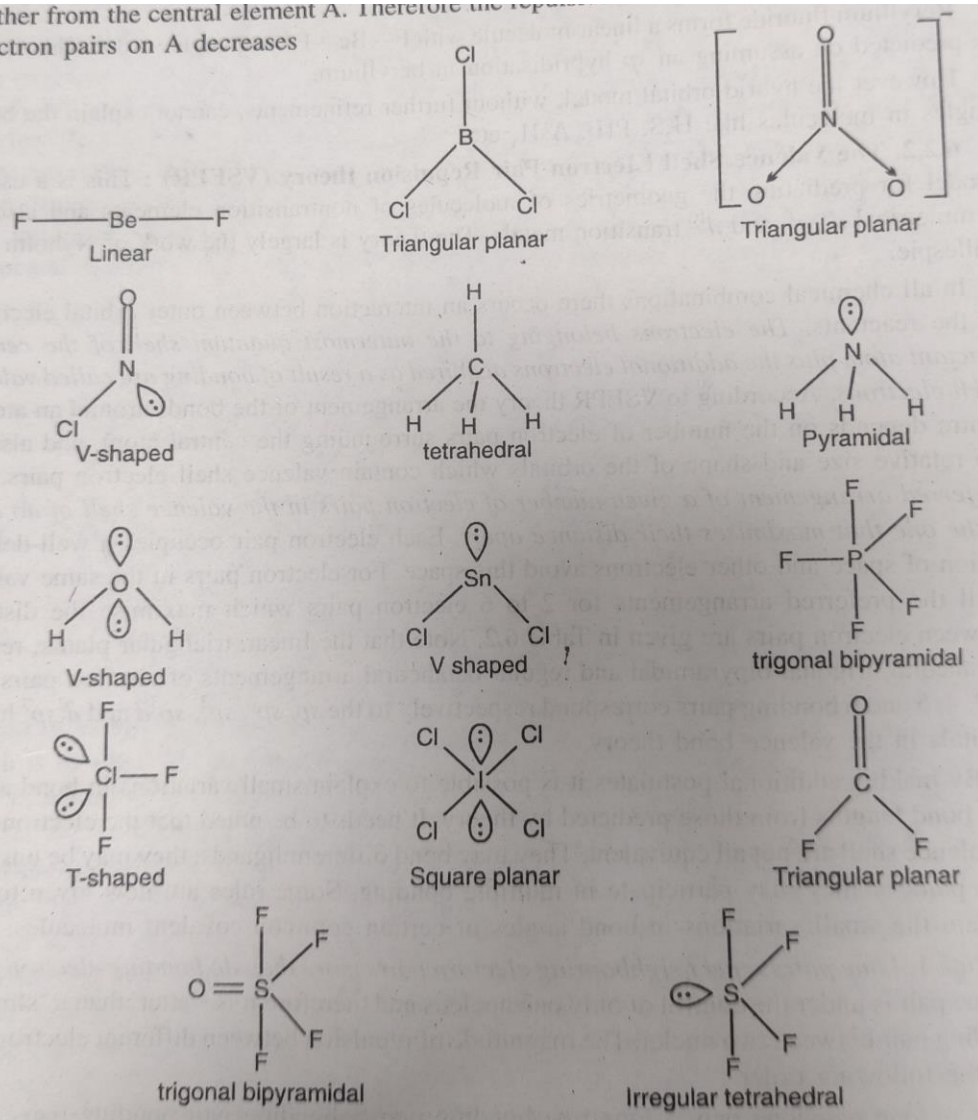


Fig. 6.8. Shapes of some molecules and ions

(Valence-shell electron-pair repulsion theory) VSEPR theory: predicting molecular geometries.

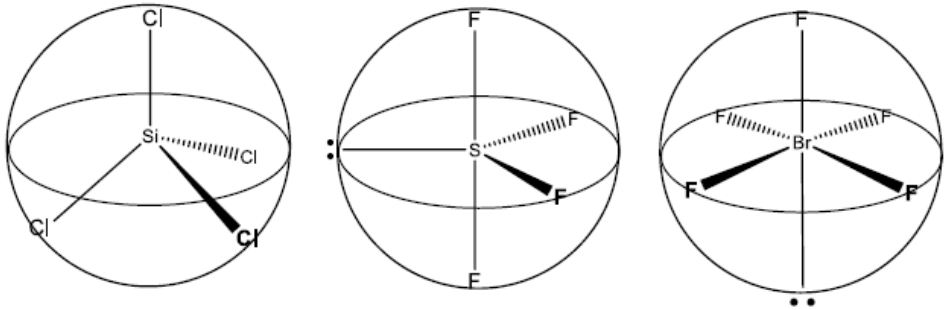
In all chemical combinations there occurs an interaction between outer orbital electrons of the reactants. The electrons belonging to the outermost quantum shell of the central reactant atom plus the additional electrons acquired as a result of bonding are called valence shell electrons. According to VSEPR theory the arrangement of the bonds around an atomic centre depends on the number of electron pairs surrounding the central atom, and also on the relative size and shape of the orbitals which contain valence shell electron pairs. The preferred arrangement of a given number of electron pairs in the valence shell of an atom is the one that maximises their distance apart. **Each electron pair occupies a well-defined region of space and other electrons avoid this space. Note that the linear, triangular planar, regular tetrahedral, trigonal bipyramidal and regular octahedral arrangements of electron pairs with 2, 3, 4, 5 and 6 bonding pairs (STERIC NUMBER).** By making additional postulates it is possible to explain small variations in bond angles and bond lengths from those predicted by theory. It needs to be noted that the electron pairs in valence shell are not all equivalent. They may bond different ligands, they may be unshared lone pairs or they may participate in multiple bonding. Some rules are now given to help explain the small variations in bond angles of certain common covalent molecules.

Remember while calculating STERIC NUMBER

The electrons are arranged in pairs that are either bonding (shared) pairs [X] or non bonding (unshared) pairs [E]. In the usual σ - π treatment this usually means ignoring the π bonds temporarily since they will follow the σ bonds and therefore the two bonding pairs of a double bond or three bonding pairs of a triple bond are all primarily considered as a single point on the sphere.

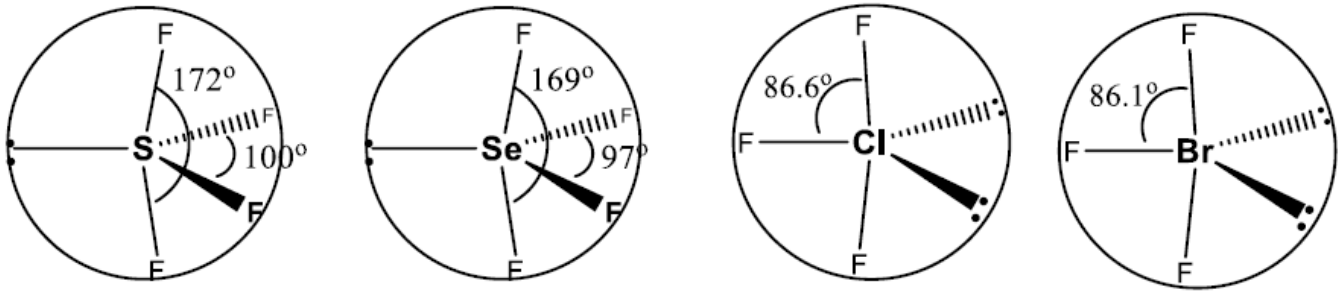
The rules

The pairs of electrons in the valence shell adopt that arrangement which maximizes their distances apart. This in other words means that the bonding and non bonding electron pairs are arranged as far apart from each other as possible on the surface of a sphere with the atom A at the centre of the sphere.

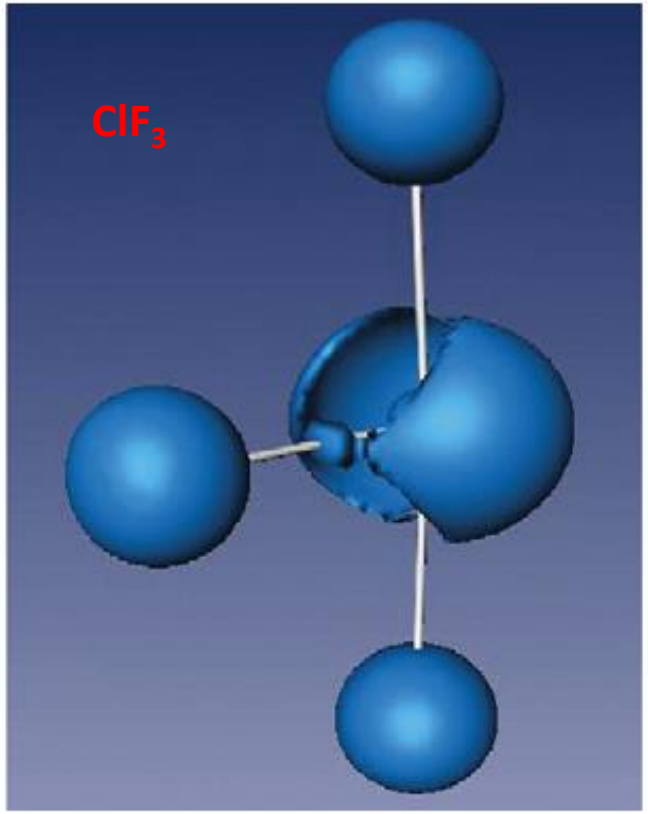


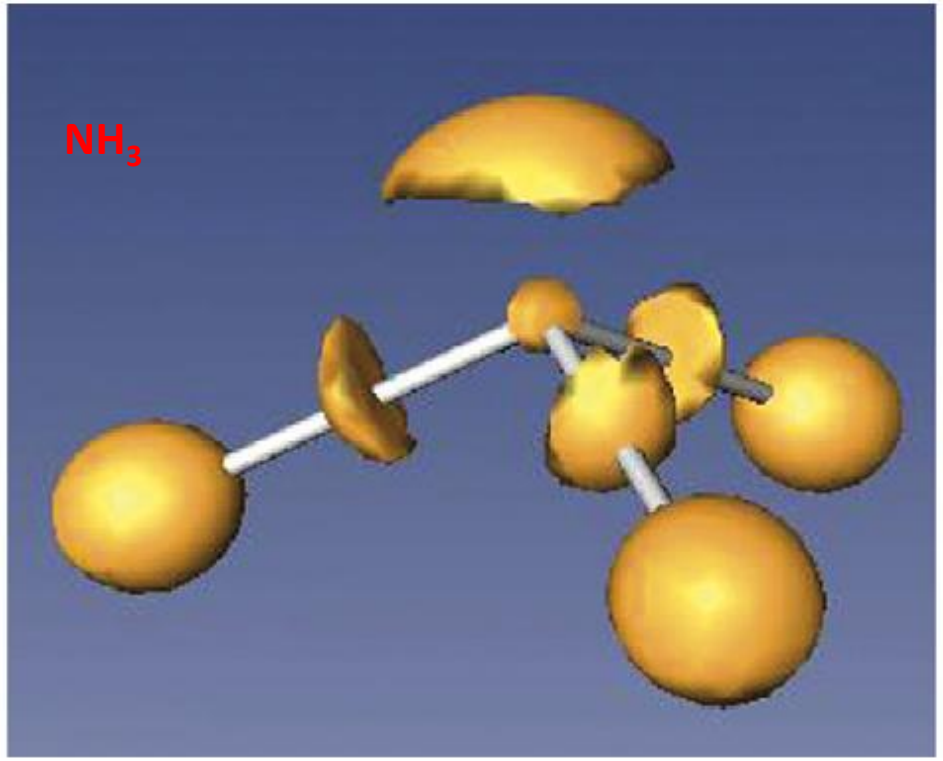
Rules which take care of the deviations observed from the ideal geometry

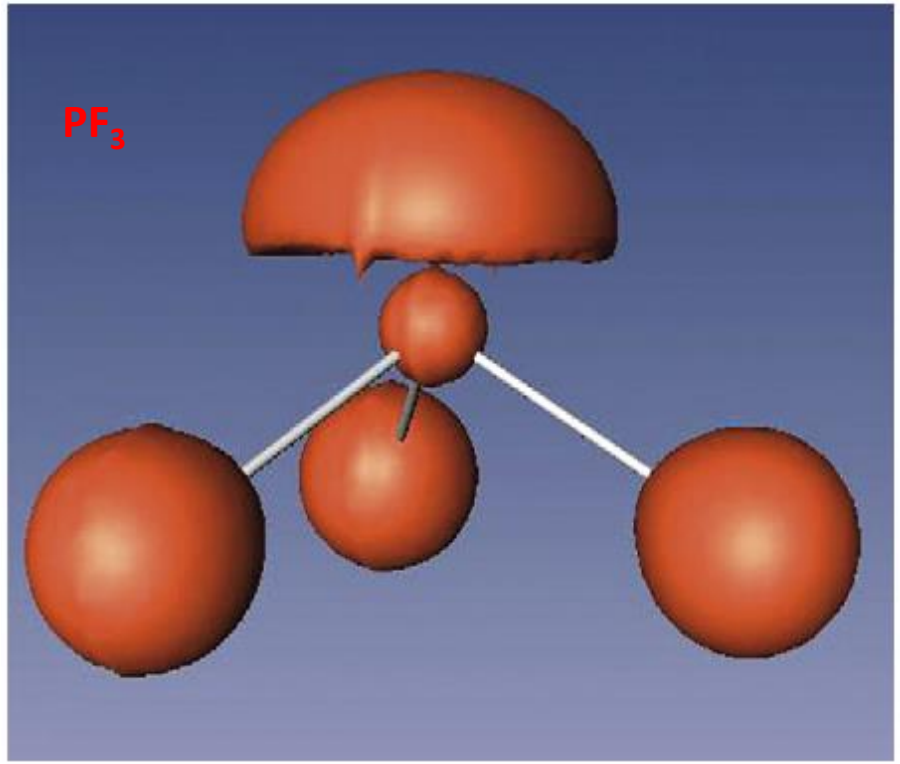
Rule 1: A non bonding lone pair [E] occupies more space on the surface of the central atom than a bonding pair [X]. In other words, lone pair- lone pair repulsion is greater than lone pair- bond pair repulsion which is greater than bond pair-bond pair repulsion. Lone pairs choose the largest site, e.g., **equatorial in trigonal bipyramid and axial in pentagonal bipyramid.** If all sites are equal, lone pairs will be better placed trans to each other.



CH_4 (109°)
NH_3 (107°)
H_2O (104°)







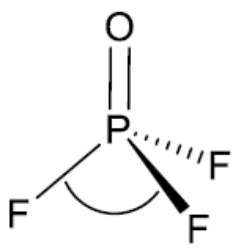
(a) ClF_3 – showing the charge concentrations corresponding to the three F atoms and the two charge concentrations corresponding to the two lone pairs in the valence shell of the central Cl atom.

(b) NH_3 – showing the nitrogen lone pair, three bonding pairs and the three H atoms.

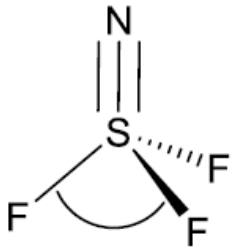
(c) PF_3 – showing the phosphorus lone pair and the three F atoms – no bonding charge concentrations are observed because of the large difference in electronegativity between P and F.

R. J. Gillespie, *Coord. Chem. Rev.* **2008**, *252*, 1315–1327.

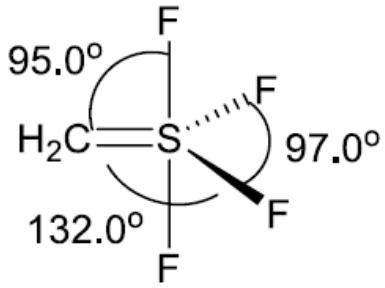
Rule 2: The strengths of the repulsions between bonds of different multiplicity decreases in the order: triple–double > double– double > double–single > single–single.

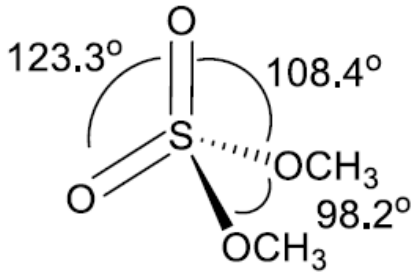


99.6°



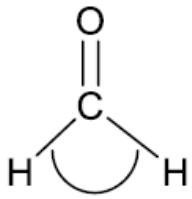
94.0°



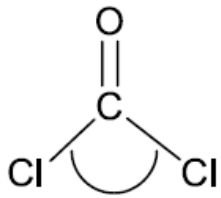


Rule 3: Bonding pairs to electronegative substituents occupy less space than those to more electropositive substituents. Therefore angles between the single bonds, decrease with increasing electronegativity of the substituent.

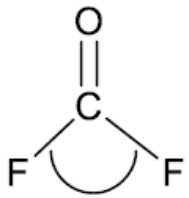
This decrease in domain size with increasing electronegativity of the substituent happens as more of the bonding electron density is shifted to the electronegative substituent. In other words, the repulsion between single bonds decreases with increasing electronegativity of the substituent and/or decreasing electronegativity of the central atom.



116°



111°



108°

107°

102°

Why sharing of electrons give rise to the formation of a stable bond?

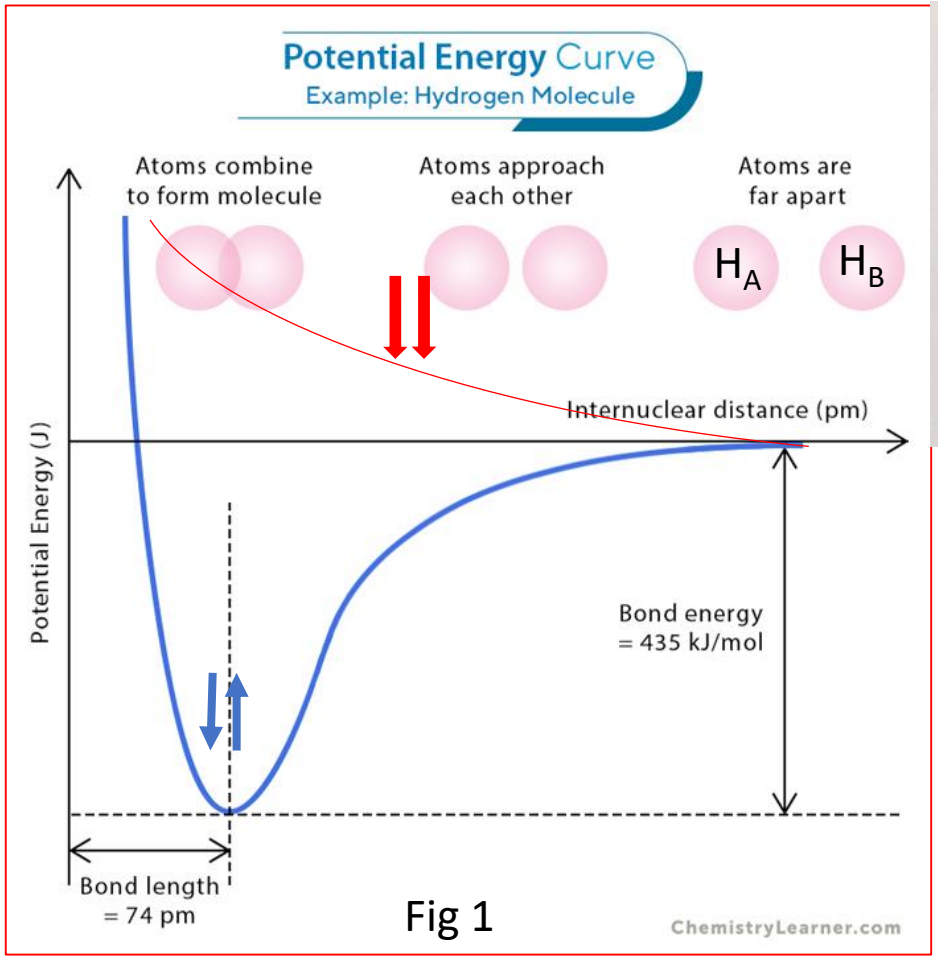


Fig 1

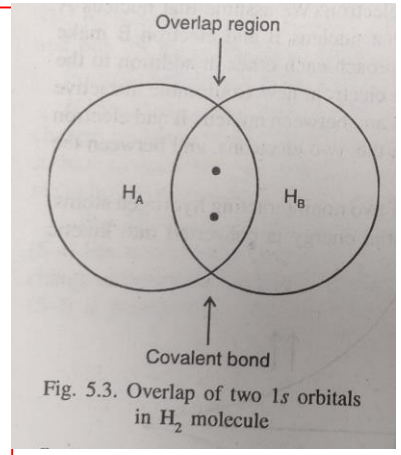


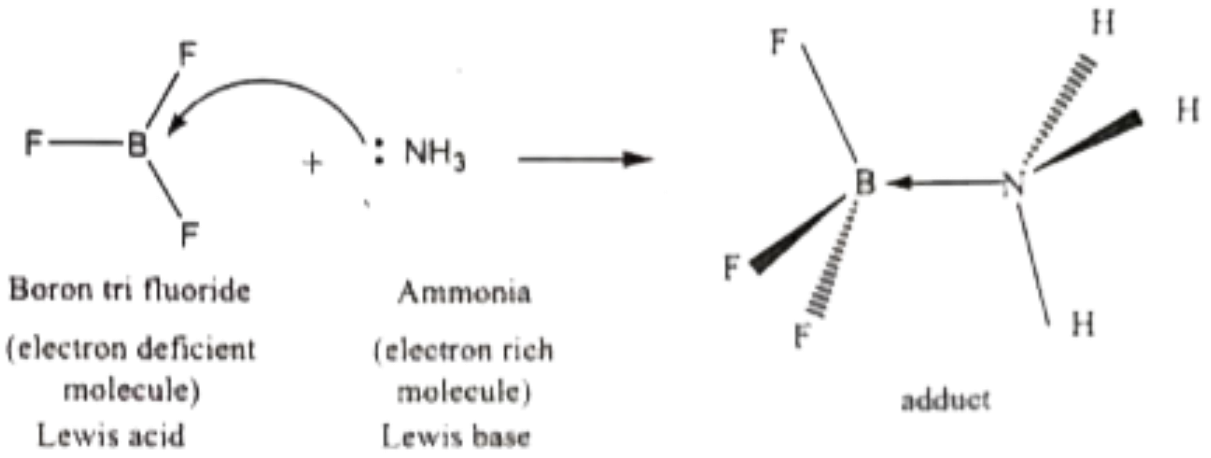
Fig. 5.3. Overlap of two 1s orbitals in H_2 molecule

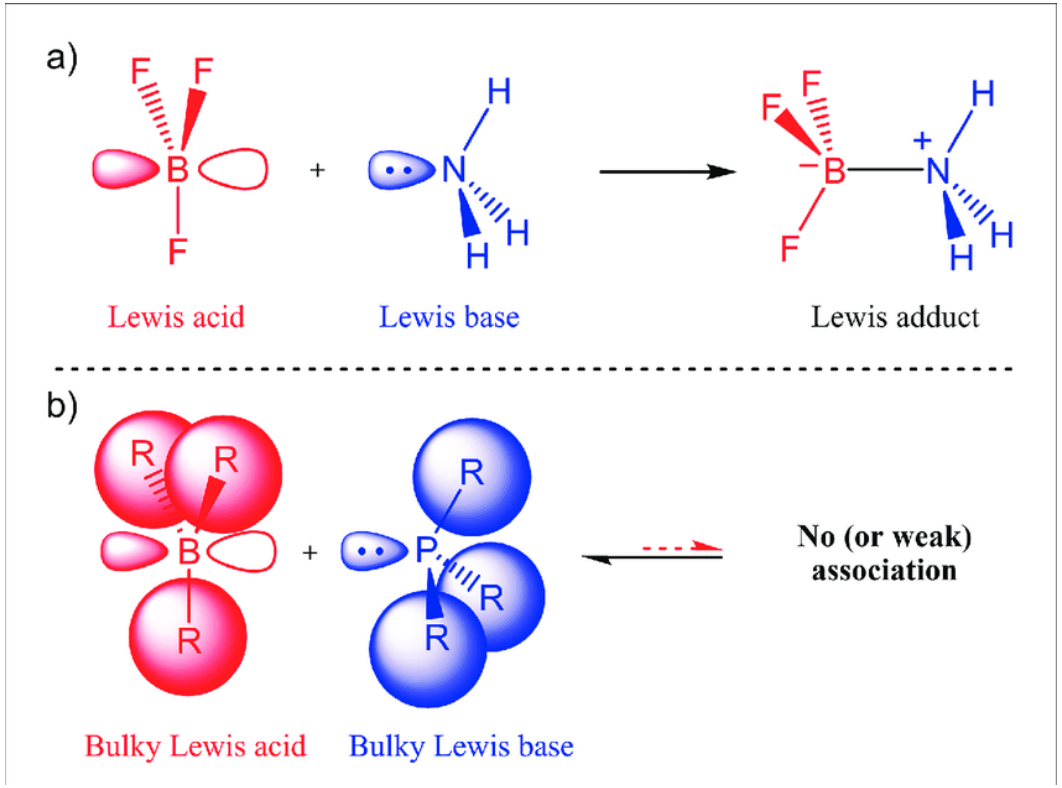
Let us consider the covalent bonding in the diatomic hydrogen molecule. Each hydrogen atom consists of a one-proton nucleus and a single 1s electron. We assume that nucleus A and electron A constitute the hydrogen atom H_A, and that nucleus B and electron B make up the second hydrogen atom H_B. As the two atoms approach each other in addition to the Coulombic attraction between each nucleus and its own electron, new Coulombic attractive forces begin to operate between nucleus A and electron B and between nucleus B and electron A. Again Coulombic repulsive forces also exist between the two electrons, and between the two nuclei.

At the optimum internuclear distance of 0.74 A (74 pm) each nucleus exerts an attractive force on its own electron as also on the electron of the other atom. Consequently the two electrons forming the covalent bond are shared by the two atoms. Quantum mechanically we say that there is a very high probability of finding the pair of electrons in between the two atoms. The higher the probability the stronger is the bond. That is, the greater is the energy necessary to break the bond. In terms of orbitals of the two electrons, there occurs a significant overlap of the two 1s orbitals of the two hydrogen atoms, such that one electron originally confined to a single orbital can now spread itself over the other orbital (Fig. 5.3). Thus the two electrons of H and H, now have a common orbital and allowing according to Pauli principle no more than two electrons per orbital, it follows that the electron spins of H_A, and H_B have to be opposite if they are to share a common orbital. In the case of two hydrogen atoms with their electron spins parallel there is no minimum in the potential energy curve (Fig. 1) i.e. no H, molecule is formed. Therefore in order that a covalent bond may be formed between two atoms a necessary condition is that each of the atoms must possess an unpaired spin.

This general approach which emphasizes overlap of atomic orbitals of individual atoms is referred to as Valence Bond (or Atomic Orbital) method. Covalent bond formation thus can occur be- tween two hydrogen atoms, between two fluorine at- oms, and between one hydrogen atom and a fluorine atom since each fulfils the criterion of having one unpaired electron in its outer orbital. But no such molecule formation can take place between two helium atoms since helium has a configuration of $1s^2$ that is, a filled 1s shell with no unpaired spin.

Concept of Lewis model, coordinate bond and steric bulk





$[CH_3]^-$ and $AlCl_3$

Concept of acid and base

(1) Predict the shape of HCN molecule

Step1: Carbon is the central element, based on Lewis model predicts the bond between C and H, and also for C and N
Step 2: Check if there is any lone pair and find out the number of sigma bond
Step 3: Predicts the shape of HCN

(2) Justify the tetrahedral shape of $[BH_4]^-$

Hints: Anion indicate one electron extra to the central atom

(3) Predicts the shapes of chlorate ion (ClO_3^-)

(4) Predicts the shape of PF_5